

# NOVA IT

## Keeping Jira as the Single Source of Truth

How Nova IT integrates with Vertex's device lifecycle, MDM stack, and compliance requirements

# The Challenge

Vertex needs Nova IT to manage hardware globally — but Jira Asset Management must stay the authoritative record for every device.

## 01

### Global Procurement

Vertex expects Nova to handle device purchasing across multiple regions. Ordering, sourcing, and logistics originate in Nova.

## 02

### Jira Stays Master

All device records must be reflected in Jira Asset Management — the system auditors query during compliance reviews.

## 03

### MDM Already in Place

Kandji (Mac) and Intune (Windows) are live. Any solution must complement them — not replace them.

# Vertex's Current Tech Stack

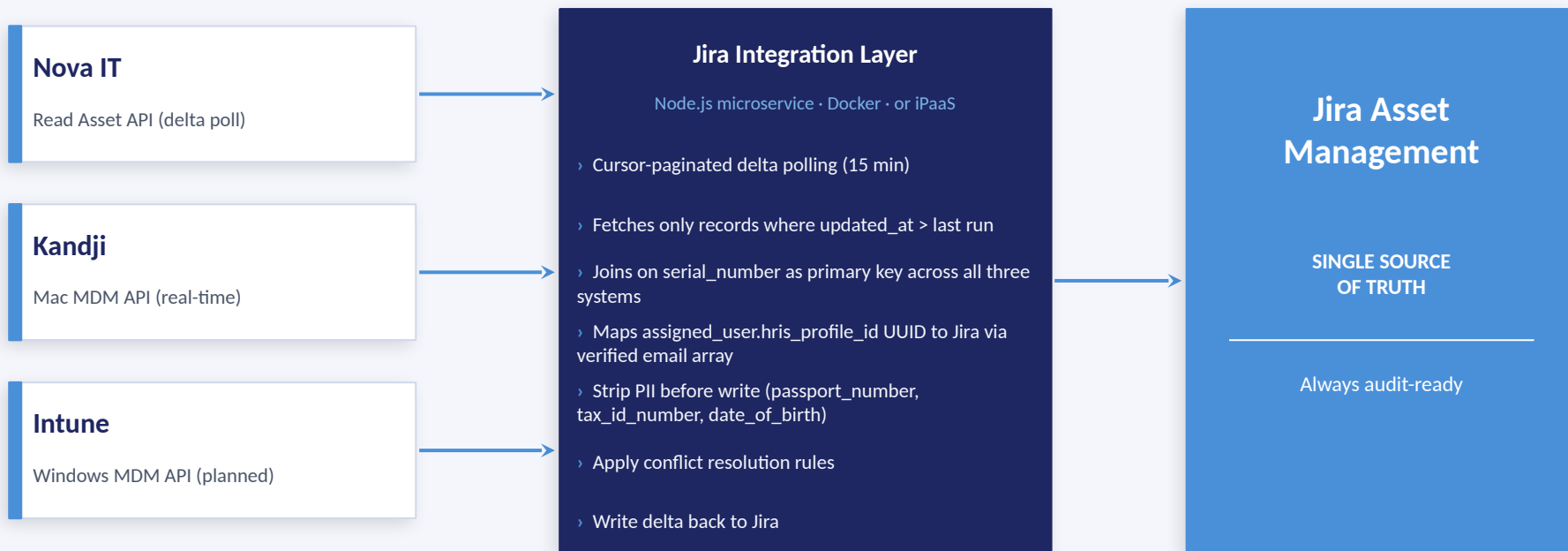
Before adding Nova, understanding what's already connected — and what isn't — is essential.

| Component             | Function                                                                                                | Integration Status          |
|-----------------------|---------------------------------------------------------------------------------------------------------|-----------------------------|
| Kandji                | Mac MDM — fleet management, OS updates, app deployment for all macOS devices                            | ✓ LIVE<br>Connected to Jira |
| Intune                | Windows MDM — policy enforcement, compliance checks, device enrollment for all Windows devices          | PLANNED<br>Roadmap          |
| Jira Asset Management | Single Source of Truth (SSOT) — the authoritative system of record queried during all compliance audits | AUTHORITATIVE               |
| Nova IT               | Hardware lifecycle & global procurement — ordering, shipping, retrieval, and device condition tracking  | PROPOSED<br>This deck       |

Nova IT acts as the primary driver for lifecycle events. MDMs provide real-time device health telemetry. Jira reconciles all three.

# BAU Sync Architecture

Jira acts as the reconciliation hub, pulling from all three sources. Nova is authoritative on procurement and lifecycle; MDMs are authoritative on device health.



# Technical Architecture Validation

This integration blueprint was not designed in a vacuum. Field mappings, lifecycle transitions, and conflict resolution rules were pre-validated against real API constraints.

01

## Prototyping Workflow

The design was pre-validated using a functional simulation harness. The architecture reflects what the Nova API actually returns — not what we assumed it returns.

02

## Containerized Emulation

Built a local, containerized Node.js MCP bridge server in Docker to active-mock Nova's production schema — replicating pagination cursors, status enumerations, and the full assigned\_user object structure.

03

## Verification Outcome

Every field mapping, lifecycle sync transition, and conflict resolution rule has been syntactically verified against live-mocked payloads. This is a structurally validated integration specification.

*Pre-validation methodology available for engineering review upon request.*

# What Nova Exposes: The Asset API

**GET** <https://api.novait.io/rest/v2/it/assets> Scope: it-assets:read

## Key Data Fields

|                                                    |                                             |
|----------------------------------------------------|---------------------------------------------|
| <code>serial_number</code>                         | Primary join key across Jira & MDMs         |
| <code>status</code>                                | Full lifecycle state (18 values)            |
| <code>location</code>                              | WITH_USER, AT_WAREHOUSE,<br>WITH_COURIER... |
| <code>assigned_user.hris_profile_id</code>         | Link to worker identity in Nova HR          |
| <code>product.name / brand</code>                  | Make and model                              |
| <code>product.specifications</code>                | Processor, RAM, storage                     |
| <code>purchase_date / purchase_value.amount</code> | Cost & age for depreciation                 |
| <code>grade</code>                                 | Condition: NEW, A, B, C, D, F               |
| <code>updated_at</code>                            | Change detection for delta sync             |

## Asset Status States

|                                 |                                     |
|---------------------------------|-------------------------------------|
| <b>ACTIVE</b>                   | In use by employee                  |
| <b>RETURNED_TO_NOVA</b>         | Retrieved, back in chain            |
| <b>AT_WAREHOUSE</b>             | Stocked, awaiting deployment        |
| <b>WITH_COURIER</b>             | In transit — not yet received       |
| <b>BEING_REPAIRED</b>           | Sent for repair                     |
| <b>DAMAGED / BROKEN (x6)</b>    | By user, in transit, at warehouse   |
| <b>LOST_IN_TRANSIT / STOLEN</b> | Flag for audit                      |
| <b>BINNED / WRITE_OFF</b>       | End of life — archive in Jira       |
| <b>ARCHIVED</b>                 | Fully removed from active inventory |

# Field Mapping: Nova → Jira Asset

serial\_number is the stable join key across all three systems. These Nova API fields map directly into Jira Asset custom fields.

| Nova API Field          | Jira Asset Field   | Source Priority        | Notes                |
|-------------------------|--------------------|------------------------|----------------------|
| serial_number           | Asset Tag / Serial | All sources            | Primary join key     |
| status                  | Asset Status       | Nova (lifecycle)       | 18 enumerated states |
| assigned_user.emails[0] | Assigned To        | MDM (if enrolled)      | Verified email match |
| product.name + brand    | Asset Name         | Nova                   | Make + model         |
| product.specifications  | Hardware Specs     | Nova / Device Insights | CPU, RAM, storage    |
| purchase_date           | Purchase Date      | Nova                   | For depreciation     |
| purchase_value.amount   | Asset Value        | Nova                   | Cost tracking        |
| grade                   | Condition          | Nova                   | NEW, A-F scale       |
| location                | Location Status    | Nova                   | Physical custody     |
| MDM: last_check_in      | Last Seen Online   | Kandji / Intune        | Device health signal |
| MDM: os_version         | Operating System   | Kandji / Intune        | Patch compliance     |

⚠ PII FILTER: passport\_number, tax\_id\_number, date\_of\_birth are stripped at the middleware layer before any write to Jira.

# Conflict Resolution: Who Wins?

When data sources disagree, these rules govern which value is written to Jira. The goal is audit accuracy, not last-write-wins.

| SCENARIO                                              | WINNER | REASONING                                                                                                                |
|-------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------|
| Nova: AT_WAREHOUSE<br>Intune: Active Online           | Intune | Intune has live telemetry. Nova likely hasn't processed the return yet. Jira shows 'Active' until Nova confirms receipt. |
| Nova: WITH_COURIER<br>Kandji: No device record        | Nova   | Device is in transit — MDM won't enroll until delivery. Jira shows 'In Transit' with Nova order data.                    |
| Nova: ACTIVE (User A)<br>Kandji: Enrolled to User B   | Flag   | Possible reassignment in progress. Jira logs a conflict flag; IT team resolves manually.                                 |
| Nova: BEING_REPAIRED<br>Kandji: Last seen 14 days ago | Nova   | Repair is a known, intentional state. Stale MDM check-in is expected. Jira reflects repair status.                       |
| Nova: serial_number missing<br>Kandji: serial present | Kandji | Jira uses Kandji's serial as the record key. A Nova alert is triggered for manual reconciliation.                        |



# Two Paths: Device Insights vs. API-Only

The integration works fully without Device Insights. If Vertex's security policy permits installation, Path A unlocks richer real-time data. If not, Path B is complete and audit-ready.

## PATH A

### With Device Insights

- ✓ Deploys via Kandji (Mac) or Intune (Windows) — no manual install
- ✓ OSQuery reads hardware specs, OS version, disk usage directly from the device
- ✓ Real-time asset telemetry enriches Jira beyond standard MDM data
- ✓ Heartbeat confirms physical possession — useful for compliance audits
- ✓ Works alongside Kandji & Intune, not instead of them

★ RECOMMENDED STARTING POINT

## PATH B

### API-Only

- ✓ No software installation — zero security policy risk
- ✓ Jira polls GET /rest/v2/it/assets every 15 minutes
- ✓ Covers all lifecycle states and field mapping
- ✓ Nova + Kandji/Intune MDM covers 95%+ of required Jira fields
- ✓ Can be upgraded to Path A at any time without re-architecture

★ START HERE — DAY ONE

# Getting to Day One: The Onboarding Plan

Three phases take Vertex from zero integration to a fully live, auditable Jira SSOT.

## 1

Week 1-2

### Initial Bulk Load

- › Export full inventory from Nova via GET /it/assets (cursor-paginated)
- › Bulk-import into Jira; match on serial\_number as primary key
- › Flag unmatched records for manual review before go-live

## 2

Week 3

### BAU Sync Goes Live

- › Activate the Jira polling job (15-min schedule against Nova Read API)
- › Activate existing Kandji → Jira sync (already live)
- › Monitor conflict flags for first 2 weeks; tune resolution rules

## 3

Week 4+

### Steady State & Audit Readiness

- › All Nova lifecycle events reflected in Jira within 15 minutes
- › Jira shows full history: Ordered → Shipped → Active → Returned → Retired
- › Optional: enable Device Insights via MDM for real-time telemetry (Path A)

# Phase 2: Optimization Roadmap

Phase 1 delivers a stable, polling-based SSOT on day one. These next steps make the architecture faster, tighter, and fully bidirectional.

## 1 Event-Driven Architecture

Nova features a native webhook engine. Once Phase 1 polling stabilizes, Vertex will subscribe Jira endpoints directly to live lifecycle events (e.g., `it.asset.assigned`). Drops sync latency from 15 min to near real-time. Payloads validated via HMAC-SHA256 signature check on the `x-nova-signature` header.

## 2 Native Jira Connector

A first-party Jira Assets integration in the Nova marketplace would eliminate the custom middleware layer — one-click auth, managed field mapping, built-in conflict resolution.

## 3 Write API for Lifecycle

Allow Jira to trigger Nova actions (offboard device, initiate retrieval) via API — completing the bidirectional loop without users ever touching the Nova UI.

## 4 MDM Status in Asset API

Surface Kandji/Intune `last_check_in` and enrollment status inside `GET /it/assets` (when Device Insights is enabled) — one API call for the full picture.

# What We're Proposing

## Jira stays the SSOT.

Every device Nova touches is reflected in Jira within 15 minutes. Auditors see the full picture — always.

## API-first, no security risk.

We recommend the API-only path on day one. No agent, no installation policy debate. Upgrade to Device Insights later when approved.

## Three sources, one record.

Nova owns procurement & lifecycle. Kandji & Intune own device health. Jira reconciles them into a single, conflict-resolved asset record.

## Built to evolve.

When Intune → Jira goes live, it slots right in. Phase 2 transitions to Nova's native webhook engine — replacing polling with near real-time sync, zero re-architecture.

## Operational ROI from day one.

One source of truth eliminates duplicate asset records, manual reconciliation, and audit prep overhead — critical at 6,500 seats across multiple geographies.

**Next Steps: Confirm field mapping with Vertex's Jira admin • Agree on conflict resolution rules • Kick off Phase 1 bulk load**